LangChain 2-Week Course — Day 1 Record

Topic: Python Setup, Environment Configuration & First LLM Call | Status: Completed & Verified

1. Day 1 Overview & Key Objectives

Core Objective: LangChain ka basic concept aur architecture samajhna, Windows par virtual environment configure karna, OpenAI API Key ko safe tariqay se .env file mein manage karna, aur pehli functional Python script (`app.py`) se LLM call karna.

2. Key Terminology & Concepts Explained

Term / Concept	Detailed Explanation & Practical Example
LLM (Large Language Model)	Bohot bara AI pattern-recognition model (jaise OpenAI ka GPT-4o) jo text predict aur generate karta hai. Fast aur intelligent engine ki tarah hota hai lekin bina tools/memory ke akela app nahi bana sakta.
LangChain	Ek powerful Python/JS framework jo LLM engine ko gaari (car) ki tarah complete infrastructure (steering, wheels, fuel) deta hai taake real-world tools, memory aur external files connect ho sakein.
Environment Variables & .env	Sensitive keys (jaise API keys) ko source code se alag rakhne ka tarika. Key leak se bachat aur billing protection ke liye `python-dotenv` library se keys system environment mein load ki jati hain.
.gitignore File	Git configuration file jo batati hai ke `.env` ya `venv/` jaisi sensitive files ko GitHub par push nahi karna taake code public hone par API key steal na ho sake.
LangChain Modules	**Model I/O:** Prompts & Outputs manage karna. **Retrieval:** External files/PDFs load karna. **Memory:** Chat history yaad rakhna. **Chains:** Multiple steps ko aapas mein jorna. **Agents:** AI ko decisions lene ke tools dena.

3. Windows Virtual Environment Setup Guide

Step-by-Step Commands (Windows):

1. Virtual environment create karna: `python -m venv venv`
2. Environment activate karna: `venv\Scripts\activate` (Terminal mein `(venv)` show hoga)
3. Packages install karna: `pip install langchain langchain-openai python-dotenv`

4. Practical Implementation (Verified Code: app.py)

```
import os from dotenv import load_dotenv from langchain_openai import ChatOpenAI # 1. Environment variables ko load karna (.env file se) load_dotenv() # 2. Custom environment variable se API Key get karna api_key = os.environ.get("OPEN_API_SECRET_KEY") # 3. LangChain Model initialize karna llm = ChatOpenAI(model="gpt-4o-mini", temperature=0.7, api_key=api_key) # 4. First LLM Prompt call response = llm.invoke("LangChain kya hai aik line mein asan Urdu ya Roman Urdu mein batao.") # 5. Result Output print("AI Response:", response.content)
```

5. Student Assessment & Verification Record

Verified Responses by Student:

1. Architecture Concept: Student ne engine & car ki example se confirm kiya ke LLM akela powerful engine hai aur LangChain usay complete app banane ka framework deta hai.

2. Memory Module Role: Student explanation: *"Memory mein hum previous chat store karte hain jisse purani baatein LLM ko milti rehti hain aur woh relevant reply deta hai."*

3. API Key Security: Student explanation: *"API key direct code mein nahi rakhte taake koi chori karke use na kare, safe way `.env` file hai kyunke woh `.gitignore` mein hoti hai aur GitHub par nahi jati."*